

Yassine Kaddami

Email: yaskaddami@gmail.com

Mobile: +33 7 51 36 49 45



EDUCATION

- **ENS Paris Saclay** Paris, France
MSc MVA (Mathematics Vision and Learning) September 2024 - September 2025
Courses: Reinforcement Learning, Convex optimization and application in Machine Learning, 3D Computer Vision, Generative Models for images, Deep Learning and signal processing introduction and industrial application, algorithms for optimization and networks management, Large Language Models, Kernel Methods, Introduction to Probabilistic and Graphical Models
- **IMT Atlantique** Brest, France
Generalist Engineering Diploma August 2021 - September 2024
Major: Mathematical and computational engineering — *Minor:* Collaborative Software Development
Courses: Stochastic Modelling, Numerical Methods, Introduction to Machine Learning, Deep Learning, Natural Language Processing and Text Mining, Probability and Statistics, Advanced Methods for Software Development and Programming, Software Development Engineering, Robotic Systems Programming
- **Lycée Charlemagne** Paris, France
Intensive Preparatory Courses September 2019 - July 2021
Two-year program for competitive entrance exams to top engineering schools in France
Track 2nd Year: Intensive advanced mathematics and physics track reserved for 1st year top-performing students (MP*)
Track 1st Year: Intensive mathematics, physics, and engineering sciences track (MPSI)

EXPERIENCE

- **AI Research Intern at Yseop** Paris, France
AI agents May 2025 - November 2025
 - **Research Focus:** Investigating how to improve tool-calling AI agents powered by small language models (SLMs) for multi-step, complex task resolution. Aiming for a lightweight yet effective alternative to large-model agents.
 - **Key Contributions:**
 - * Conducted a comprehensive review of state-of-the-art research papers and open-source benchmarks on tool-calling agents.
 - * Developed a new taxonomy of tool-calling agent tasks to structure the research and identify high-impact subproblems.
 - * Experimented with basic agentic paradigms using SLMs instead of large models ($\leq 1B$ parameters), identifying recurring failure modes.
 - * Designed novel agent architecture and paradigm to address identified limitations in existing paradigms.
 - **Skills:** AI Agents, Small Language Models (SLMs), Policy Optimization, Literature Review
- **Data Scientist Intern at Cdiscount** Bordeaux, France
Generative AI R&D April 2024 - Sep 2024
 - **Automation of high-quality SEO content Generation:** Researching and innovating advanced techniques to automate the generation of high-quality marketing content using open-source Large Language Models (LLMs) while adhering to legal constraints.
 - **AI content control and evaluation:** Implementing a model to detect harmful content and defining metrics for evaluating the generated content.
 - **Skills:** LLM fine-tuning, Knowledge Distillation, RAG, Quantization, Parallel Computing, Snowflake, Ceph, SQL, literature review of state-of-the-art articles, Natural Language Processing (NLP)
- **Python Development Internship at Exail** Lannion, France
Object Oriented Programming April 2023 - August 2023
 - **Python app Development:** Improving the conception of production steps of a fiber in Exail by developing a Python app for Exail technicians using object oriented programming and following agile methodology.
 - **Skills:** external communication (direct communication with the clients, definition of their needs and continuous adaptation to them), internal communication (weekly presentation to the tutors), initiatives taking, autonomy, rapid adaptation to new contexts.
 - **Tools:** Jira, Git, BitBucket, MySQL, PyQt, Figma, SQL, class diagrams, design patterns

RESEARCH PROJECT

- **Hyperparameter Transfer for Large Neural Networks: Limiting behavior of Neural Networks:**
 - **Research Paper:** Demonstrated the unique properties of the μ Parametrization by studying the limit behavior of large multi-layer perceptrons.
 - **Skills:** Linear Algebra, Random Matrix Theory, Probability

PROJECTS

- **Speculative decoding:** Implementing paper on speculative decoding by Yaniv et.al Fast Inference from Transformers via Speculative Decoding
- **Big Data and Cloud Computing project:** Applying certain statistical methods and linear regression models to Big Data on the Google Cloud Platform using BigQuery.
- **Image Segmentation with Transformers:** Understanding TransUNet model and code, and experimenting it on new applications. Skills: Complex technical articles and codes comprehension, Adapting state-of-the-art Deep Learning models to new applications.
- **Closed Chatbot using RAG:** Implementing a chatbot responding to covid 19 scientific questions using open source corpus and models.

VOLUNTEER EXPERIENCE

- **Sports Office (BDS) IMT Atlantique**
Part of the International Pole: Organized sports events to integrate international students.
- **Casablanca Heritage Day Guide Morocco**
Attended courses on Casablanca's heritage and history and served as a guide at various historic sites during Heritage Day.

LANGUAGES

English		French		Arabic		Spanish		German
C1 Level		Native proficiency		Mother tongue		A2 Level		A2 Level
IELTS 7.5/9		French High School Diploma						